



KTU NOTES

The learning companion.

**KTU STUDY MATERIALS | SYLLABUS | LIVE
NOTIFICATIONS | SOLVED QUESTION PAPERS**

🌐 Website: www.ktunotes.in

DYNAMIC CIRCUITS - Module 2

A circuit having constant energy sources is said to be in steady state if the currents and voltages do not change with time. Also the amplitude and frequency of a sinusoidal ~~source~~ quantity never changes with time in a steady state condition.

In a network containing energy storage elements, when there is a change in the excitation, the currents and voltages change from one steady state to another steady state. This behaviour of the voltage or current when it is changed from one steady state to another steady state is called ~~the~~ transient state.

When there are energy storage elements (which are independent of the sources) the response depends upon the nature of the circuit and is called natural response. These energy storing elements deliver their energy to the resistances and hence the response changes with time and is referred to as transient response.

When we consider the sources acting in a circuit, the response depends upon the nature of the source. This response is called forced response.

The application of KVL or KCL to circuits containing energy storage elements results in ^{integro-} differential equations. The solution of differential equations has 2 parts. The particular integral and complementary function.

The complementary function dies out after a short period of time and is referred to as transient response or source free response.

The particular solution is steady state response or forced response.

To find the complete solution, a differential equation has to be formed. 1st order circuits contains only one energy storing element i.e. inductor or capacitor.

In general transient currents are produced whenever

1. An apparatus or circuit is suddenly connected or disconnected from the supply.
2. The circuits are short circuited &
3. There is a sudden change in the applied voltage from one finite value or another.

Thus whenever circuits are suddenly switched on or off from the supply voltage, the resultant current consists of two parts.

1. A final steady state current or normal current &
2. A transient current superimposed on the steady state current.

The transient currents are not driven by any part of the applied voltage, but are entirely associated with the changes in the stored energy in inductance and capacitance. Since there is no stored energy in resistors, there are no transients in a purely resistive circuit. $\therefore$ If an emf is applied suddenly to a pure resistance, the current rises abruptly or instantly to its steady value. If the same emf is applied to an inductive coil, the steady state value is reached only after some time. This is because, time is required to store energy in the magnetic field.

Types of transients - There are single energy transients and double energy transients.

Single energy transients are those in which only one form of energy either electromagnetic or electrostatic is involved as in RL and RC circuits.

Double energy transients are those in which both electromagnetic and electrostatic energies are involved as in R-L-C circuit.

Though transient periods are of short duration, it is during this period that some of the most serious problems are encountered. It doesn't mean that transients are undesirable, some circuit operations depends on such transient phenomena.

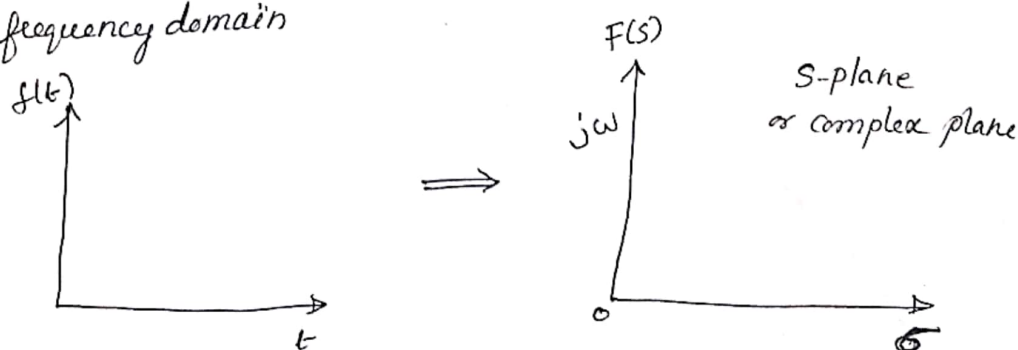
Transient Analysis using Laplace transforms

In electric circuits, we often come across integro-differential equations especially with energy storage elements. Hence it is more convenient to adopt the Laplace transform technique for solving such equations.

Advantages of Laplace transform technique.

1. It simplifies exponential and trigonometric functions into simple algebraic functions [eg. $\mathcal{L}[e^{at}] = \frac{1}{s-a}$ & $\mathcal{L}[\sin \omega t] = \frac{\omega}{s^2 + \omega^2}$]
2. It transforms differentiation and integration respectively into multiplication and division. [eg. $\mathcal{L}\left[\frac{df}{dt}\right] = sF(s) - f(0^+)$
 $\mathcal{L}\left[\int f(t) dt\right] = \frac{F(s)}{s}$]
3. In solving diff. equation by this technique, the solution comes out complete. The arbitrary constants need not be determined separately. On the other hand, if we use ordinary classical method, the complementary function and particular integral have to be determined. Finally the arbitrary constants have to be determined from the known initial conditions. Thus the Laplace transform technique gives an easier method for circuit analysis.

Laplace transform transforms time domain functions to frequency domain



' s ' is the complex variable $s = \sigma + j\omega$, where ' σ ' represents frequency & ω is the radian frequency.

Laplace Transform of a function $f(t)$ is defined as

$$F(s) = \mathcal{L}[f(t)] = \int_0^{\infty} f(t) e^{-st} dt$$

The variable 's' is the Laplace operator, which is a complex variable $s = \sigma + j\omega$ (transform time domain to frequency domain)

Integrodifferential equations are converted into simple algebraic equations using Laplace transform. Solution of algebraic equations are simple when compared to the solution of differential equations and s-domain solution is obtained. But for practical purpose, we require the response in time domain. So time domain response is obtained by taking the inverse Laplace transform of s-domain response.

Inverse Laplace transform

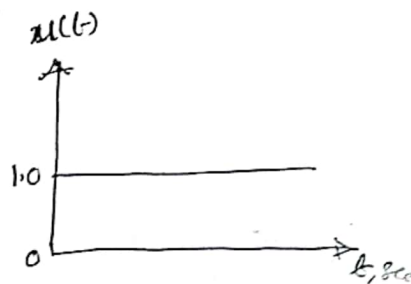
$$f(t) = \mathcal{L}^{-1}[F(s)] = \frac{1}{2\pi j} \int_{\sigma - j\infty}^{\sigma + j\infty} F(s) e^{st} ds$$

① Laplace transform of unit step $u(t)$

$$F(s) = \mathcal{L}[u(t)]$$

$$= \int_0^{\infty} e^{-st} f(t) dt$$

$$= \int_0^{\infty} e^{-st} u(t) dt = \int_0^{\infty} e^{-st} dt = \left[\frac{e^{-st}}{-s} \right]_0^{\infty} = \frac{e^{-\infty}}{-s} - \frac{e^0}{-s} = \frac{1}{s}$$



② Exponential function of t

$\mathcal{L}[e^{at}]$, where 'a' is a constant.

$$\begin{aligned}\mathcal{L}[e^{at}] &= \int_0^{\infty} (e^{at}) e^{-st} dt = \int_0^{\infty} e^{-(s-a)t} dt \\ &= \underline{\underline{\frac{1}{s-a}}}\end{aligned}$$

$$\textcircled{3} \quad \mathcal{L}[e^{-at}] = \frac{1}{s+a}$$

$$\begin{aligned}\textcircled{4} \quad \mathcal{L}[\sin \omega t] &= \frac{1}{2j} \left[\mathcal{L}[e^{j\omega t} - e^{-j\omega t}] \right] \\ &= \frac{1}{2j} \left[\frac{1}{s-j\omega} - \frac{1}{s+j\omega} \right] \\ &= \frac{1}{2j} \left[\frac{s+j\omega - (s-j\omega)}{s^2 + \omega^2} \right] \\ &= \frac{1}{2j} \left[\frac{j2\omega}{s^2 + \omega^2} \right] = \frac{\omega}{s^2 + \omega^2}\end{aligned}$$

$$\begin{aligned}5. \quad \mathcal{L}[\cos \omega t] &= \frac{1}{2} \left[\mathcal{L}\left[\frac{e^{j\omega t} + e^{-j\omega t}}{2} \right] \right] \\ &= \frac{1}{2} \left[\frac{1}{s-j\omega} + \frac{1}{s+j\omega} \right] \\ &= \frac{1}{2} \left[\frac{s+j\omega + s-j\omega}{s^2 + \omega^2} \right] = \underline{\underline{\frac{s}{s^2 + \omega^2}}}\end{aligned}$$

$$6 \quad \mathcal{L}[t^n] = \frac{n!}{s^{n+1}}$$

$$I. \quad \mathcal{L}[e^{-at} f(t)] = F[s+a]$$

Function, $f(t)$	Laplace transformed function, $f(s)$
1. Constant, E	E/s
2. e^{at} , e^{-at}	$\frac{1}{s-a}$, $\frac{1}{s+a}$
3. $\sin \omega t$	$\frac{\omega}{s^2 + \omega^2}$
4. $\cos \omega t$	$\frac{s}{s^2 + \omega^2}$
5. t , t^n	$\frac{1}{s^2}$, $\frac{n!}{s^{(n+1)}}$
6. $t e^{-at}$	$\frac{1}{(s+a)^2}$
7. $e^{-at} \sin \omega t$	$\frac{\omega}{(s+a)^2 + \omega^2}$
9. $e^{-at} \cos \omega t$	$\frac{(s+a)}{(s+a)^2 + \omega^2}$
10. $\frac{df(t)}{dt}$	$sF(s) - f(0^+)$
11. $\int f(t) dt$	$\frac{F(s)}{s}$
12. $\sinh \omega t$	$\frac{\omega}{s^2 - \omega^2}$
13. $\cosh \omega t$	$\frac{s}{s^2 - \omega^2}$

If the Laplace transformed function $F(s)$ is given, to find the original function $f(t)$, we have to find the inverse Laplace transform.

For eg. $F(s) = \frac{s}{s^2 + 5s + 6} = \frac{s}{(s+3)(s+2)}$

Expand into partial fractions and find the unknown coefficients A & B .

$$\text{Thus } F(s) = \frac{A}{s+3} + \frac{B}{s+2}$$

$$= \frac{3}{s+3} + \frac{-2}{s+2}$$

$$f(t) = \mathcal{L}^{-1}[F(s)] = \underline{\underline{3e^{-3t} - 2e^{-2t}}}$$

$$A = \frac{-3}{-3+2} = \underline{\underline{3}}$$

$$+ B = \frac{-2}{-2+3} = \underline{\underline{-2}}$$

$$A = \frac{s}{s+2} \bigg|_{s=-3} = \frac{-3}{-3+2} = \underline{\underline{3}}$$

$$B = \frac{s}{s+3} \bigg|_{s=-2} = \frac{-2}{-2+3} = \underline{\underline{-2}}$$

DC Transients (L-T Method)

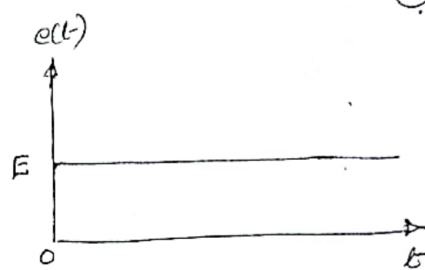
Step response of R-L circuit

(Forced Response)

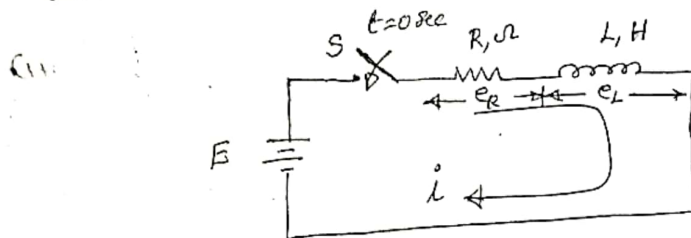
Step function is defined as

$$e(t) = 0 \text{ for } t < 0$$

$$= E \text{ for } t \geq 0$$



Step function is used to represent the closing of a circuit connected to a constant voltage system.



The RL circuit is connected to the voltage source E through a switch 'S'.

Assume that the switch is closed at $t = 0$ sec and also assume that at the time of switching, the current

$$i(0) = 0$$

Applying KVL to the circuit, we get

$$Ri + L \frac{di}{dt} = E$$

This is a differential

equation in i and to get a solution for i , apply Laplace transform technique. Taking Laplace transforms on both sides of the equation, we get

$$RI(s) + L[SI(s) - i(0)] = \frac{E}{s}, \text{ where } L[i'(t)] = I'(s)$$

Just before closing the switch $i(0) = 0$, $\therefore$ The equation reduces to

$$RI(s) + L \cdot SI(s) = E/s$$

$$I(s)[R + Ls] = E/s$$

$$I(s) = \frac{E}{s(R + Ls)} = \frac{E/L}{s(s + R/L)}$$

Expanding into partial fractions and taking inverse laplace transform, we will get the solution for $i(t)$

$$\frac{E/L}{s(s+R/L)} = \frac{A}{s} + \frac{B}{(s+R/L)}$$

$$A = \frac{E/L}{s+R/L} \Big|_{s=0} = \frac{E}{L} \times \frac{L}{R} = \frac{E}{R}$$

On solving, $A = E/R$ & $B = -E/R$

$$B = \frac{E/L}{s} \Big|_{s=-R/L} = \frac{E/L}{-R/L} = -E/R$$

$$\therefore I(s) = \frac{E/R}{s} - \frac{E/R}{(s+R/L)}$$

$$\therefore i(t) = \frac{E}{R} (1 - e^{-R/L t})$$

As $t \rightarrow \infty$, $i(t) = E/R$. $\therefore$ The final steady state component of the resultant current is $I_s = E/R$ and the transient component of the current $I_t = -\frac{E}{R} e^{-R/L t}$

$L/R = \tau$ is called the timeconstant of the RL circuit

$$\text{At } t = \tau = L/R, i(\tau) = \frac{E}{R} (1 - e^{-1}) = 0.632 E/R$$

$\therefore$ Time constant of the RL circuit is defined as the time taken to reach 63.2% of the final steady state value.

Transient voltage across R & L

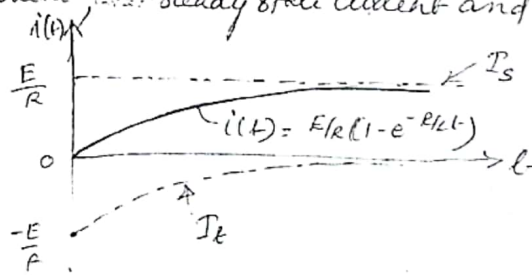
$$e_R = Ri = \frac{E}{R} (1 - e^{-t/\tau}) \times R = E (1 - e^{-R/L t})$$

$$e_L = L \cdot \frac{di}{dt} = L \cdot \frac{E}{R} \left[-e^{-R/L t} \times (-R/L) \right] = \underline{\underline{E e^{-R/L t}}}$$

$$\underline{P_R = e_R \cdot i \quad \& \quad P_L = e_L \cdot i}$$

Variation of transient current, steady state current and resultant current with time is plotted as shown.

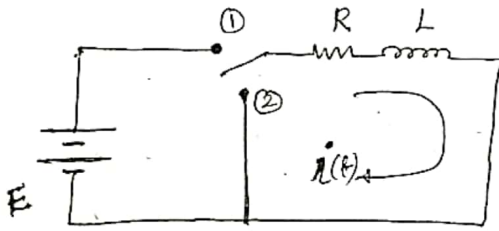
Transient current decays exponentially. $i(t) = I_s + I_t$



Transients in shortcircuited RL circuit (Natural Response)

The response of the circuits when no sources or forcing functions are present is termed as the natural response because it depends only on the nature of the circuit i.e. the type of the elements, their sizes and the interconnection of these elements. This response arises from the presence of initial energy storage within the inductive or capacitive elements in the circuit.

Natural response of RL circuit



Consider the RL circuit. The switch 'S' is previously ~~in position 1~~ in position ① and has been there till steady state conditions ~~are~~

are reached. The current then flowing is $i = \frac{E}{R} A$.
 $\therefore$ At $t = 0$, the switch is turned to position ② thereby short-circuiting the RL series circuit. i.e. voltage source is excluded from the circuit.

The application of Kirchhoff's voltage to the closed circuit gives

$$iR + L \cdot \frac{di}{dt} = 0 \quad , \quad \text{At } t = 0, i(0) = E/R$$

Taking Laplace transform on both sides

$$RI(s) + L[SI(s) - i(0)] = 0$$

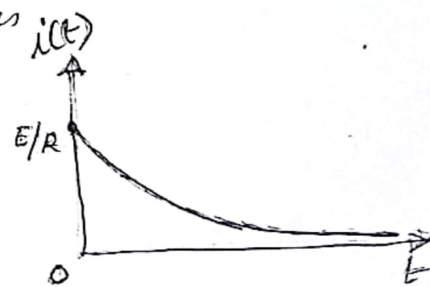
$$i(0) = E/R$$

$$\text{Then, } RI(s) + L[SI(s) - E/R] = 0$$

$$I(s)[R + Ls] = \frac{E}{R} \cdot L$$

$$I(s) = \frac{L}{R} \cdot E \cdot \frac{1}{R + Ls} = \frac{E/R}{(s + R/L)}$$

$$i(t) = \frac{E}{R} e^{-R/L t}$$



A series RL circuit with $R=100\Omega$ and $L=20H$ has a dc voltage of $200V$ through a switch at $t=0\text{ sec}$.

- Find a) the equation for the current and voltages across different elements. b) the current at $t=0.5\text{ sec}$
c) the current at which $e_R = e_L$

Solution

$$a) i = \frac{E}{R}(1 - e^{-R/L t}) = \underline{2(1 - e^{-5t})}$$

$$e_R = iR = \underline{200(1 - e^{-5t})}$$

$$e_L = L \cdot \frac{di}{dt} = 20 \times 2 \times (-e^{-5t}) \times 5 = \underline{200e^{-5t}}$$

$$b) i(t) = i(0.5\text{ sec}) = 2(1 - e^{-5 \times 0.5}) = \underline{1.836 A}$$

$$c) e_R = e_L$$

$$200(1 - e^{-5t}) = 200e^{-5t}$$

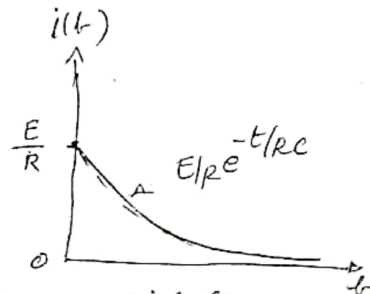
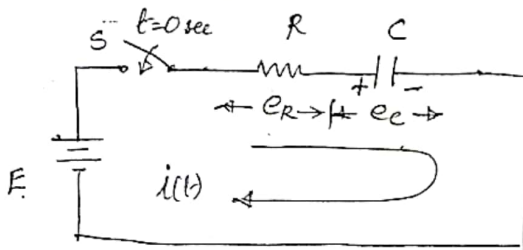
$$1 - e^{-5t} = e^{-5t}$$

$$2e^{-5t} = 1, e^{-5t} = 0.5, t = \underline{0.1386 \text{ sec}}$$

$$\therefore i = 2(1 - e^{-5 \times 0.1386}) = \underline{1 \text{ Amp}}$$

Transients in RC circuit

Step response of RC circuit



Consider the RC circuit with a DC voltage applied through a switch 'S'.

If the capacitor has an initial charge of Q_0 Coulombs, then the voltage across the capacitor immediately after closing the switch is, $V_0 = \frac{Q_0}{C}$.

Here let us assume that the capacitor is initially uncharged.

$$\therefore V_0 = 0.$$

Applying KVL to the circuit, we get

$$iR + \frac{1}{C} \int i dt = E \quad \text{This is an integral equation}$$

in 'i' and to find a solution for i, take Laplace transforms on both sides

$$R I(s) + \frac{1}{C} \cdot \frac{I(s)}{s} = E/s$$

$$I(s) \left[R + \frac{1}{Cs} \right] = E/s$$

$$I(s) = \frac{E}{s(R + \frac{1}{Cs})} = \frac{E/R}{s + 1/RC}$$

$$\begin{aligned} &= \frac{E \cdot Cs}{s(RCs + 1)} \\ &= \frac{E/R}{s + 1/RC} \end{aligned}$$

$$\text{Then } i(t) = \mathcal{L}^{-1}[I(s)] = \underline{\underline{E/R e^{-t/RC} \text{ Amp.}}}$$

$$\text{when } t=0, \underline{\underline{i(0) = E/R}} \quad \& \quad \text{when } t=\infty, \underline{\underline{i(\infty) = 0}}$$

$\therefore$ The current decays exponentially from the initial value E/R to zero as $t \rightarrow \infty$. $\therefore$ The final steady state current is zero in this case.

Explanation - The charge across a capacitor cannot change instantaneously. Hence the abrupt change in the voltage arising from the closing of the switch at $t=0$ cannot appear across the capacitor. Hence the capacitor behaves like a short circuit initially to an abrupt change in voltage.

$\therefore$ The entire voltage E appears the resistance at $t=0$ sec.

$$\therefore i(0) = E/R \text{ itself.}$$

As time increases, voltage across the capacitor V_c increases with the polarity as shown in fig.

and at $t = \infty$, E_c becomes equal to the applied voltage ' E ' and opposite in polarity. Therefore the net voltage in the circuit is zero and hence the final steady state current becomes zero.

$$\therefore \text{At } t = \infty, i(\infty) = 0$$

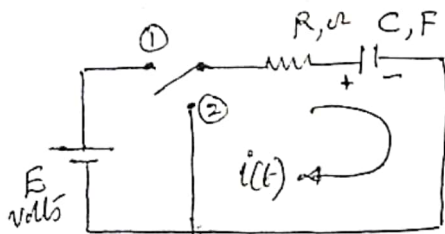
$RC = \tau$, is called the time constant of RC circuit

$$\text{At } t = \tau = RC, i(t) = \frac{E}{R} e^{-1} = 36.8 \% \text{ of initial current.}$$

Time constant of an RC circuit is defined as the time taken by the current to decay to 36.8 % of its initial value.

transient voltage across 'R'	$= iR = E e^{-t/RC}$	$\neq$
" " " " C	$= \frac{1}{C} \int i dt = E(1 - e^{-t/RC})$	
Energy stored	$= \frac{1}{2} C E^2 \text{ Joules.}$	

Transient current in a short circuited RC circuit
(Natural response)



The switch was in position (1) for a sufficiently long time and the capacitor is charged to a voltage ' E ' volts. Now at time, $t = 0 \text{ sec}$

the switch is closed to position (2), causing the capacitor to discharge through ' R '.

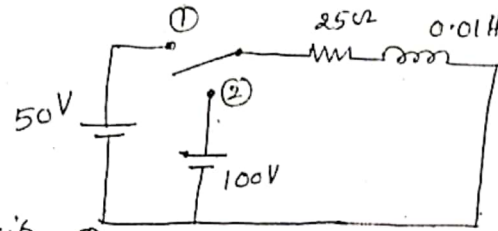
At the instant of shifting the switch to position (2), voltage across ' C ' is E .

Applying KVL, we get

$$Ri + \frac{1}{C} \int i dt + E = 0$$

$$\text{or } Ri + \frac{1}{C} \int i dt = -E$$

On the RL circuit shown below, the switch is in position ① long enough to establish steady state conditions and at $t = 0$ sec, it is turned to position 2. Find the resultant current.



Soln:

When switch is in position ① steady state conditions have been reached and hence ~~$i = \frac{V}{R} = \frac{50}{25} = 2A$~~

$$i = \frac{V}{R} = \frac{50}{25} = 2A.$$

i.e. when $t = 0$, $i(0) = \frac{50}{25} = 2A$

When 'S' is turned to position ②, 100V source is connected to RL circuit.

The equation is $25i + 0.01 \frac{di}{dt} = 100$

Taking Laplace transforms,

$$25I(s) + 0.01[sI(s) - i(0)] = \frac{100}{s}, \text{ Here } i(0) = 2A.$$

Then, $25I(s) + 0.01[sI(s) - 2] = \frac{100}{s}$

$$25I(s) + 0.01sI(s) - 2 \times 0.01 = \frac{100}{s}$$

$$I(s)[25 + 0.01s] = \frac{100}{s} + 0.02$$

$$I(s) = \frac{100}{s(25 + 0.01s)} + \frac{0.02}{(25 + 0.01s)}$$

$$= \frac{10000}{s(s + 2500)} + \frac{2}{s + 2500}$$

Expanding $\frac{10000}{s(s + 2500)}$ into partial fractions

$$\frac{10^4}{s(s + 2500)} = \frac{A}{s} + \frac{B}{s + 2500}, \text{ on solving}$$

$$A = 4 \quad \& \quad B = -4.$$

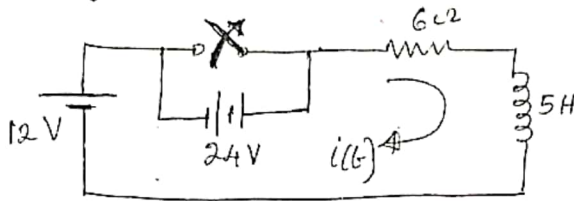
Then $I(s)$ becomes

$$I(s) = \frac{4}{s} - \frac{4}{(s+2500)} + \frac{2}{s+2500}$$

$$= \frac{4}{s} - \frac{2}{s+2500}$$

Then $i(t) = (4 - 2e^{-2500t})$ Amp.

- ③ Find the current in the circuit shown in fig. at an instant 't' after opening the switch if a current of 1 amp. had been passing through the circuit at the instant of opening.



After 't' seconds of opening the switch, applying KVL, we get

$iR + L \frac{di}{dt} = E$, where E is the total voltage of the circuit after opening the switch, i.e. $E = 12 + 24 = 36V$.

Given that, $i(0) = 1A$. Substituting for R, L & E,

$$6i + 5 \frac{di}{dt} = 36 \quad \text{Taking Laplace transforms}$$

$$6I(s) + 5[sI(s) - i(0)^+] = \frac{36}{s}$$

$$I(s)(6 + 5s) = \frac{36}{s} + 5 \times 1$$

$$= \frac{36 + 5s}{s}$$

$$I(s) = \frac{5s + 36}{s(5s + 6)} = \frac{A}{s} + \frac{B}{5s + 6}$$

$$A = \left. \frac{5s + 36}{5s + 6} \right|_{s=0} = \frac{36}{6} = \underline{\underline{6}}$$

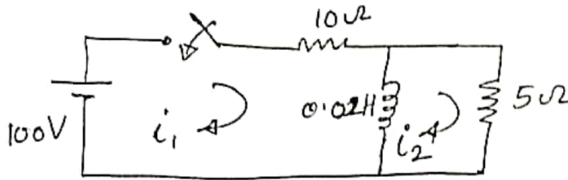
$$B = \left. \frac{5s + 36}{s} \right|_{s = -\frac{6}{5}} = \frac{30 + 36}{-6} = \underline{\underline{-25}}$$

$$\therefore I(s) = \frac{6}{s} - \frac{25}{5s+6} = \frac{6}{s} - \frac{5}{s+1.2}$$

$$i(t) = \mathcal{L}^{-1}[I(s)]$$

$$= \underline{\underline{6 - 5e^{-1.2t}}}$$

- ④ In the network, find the currents i_1 & i_2 that results, when the switch is closed at $t=0$ sec. Assume that $i(0)=0$.



The loop equations for the network are

$$10i_1 + 0.02 \frac{di_1}{dt} - 0.02 \frac{di_2}{dt} = 100 \quad \text{--- (1)}$$

$$0.02 \frac{di_2}{dt} + 5i_2 - 0.02 \frac{di_1}{dt} = 0 \quad \text{--- (2)}$$

Taking Laplace transforms on both sides of these equations eqn (1) becomes

$$10I_1(s) + 0.02 [sI_1(s) - i_1(0)] - 0.02 [sI_2(s) - i_2(0)] = \frac{100}{s}$$

$$10I_1(s) + 0.02 s [I_1(s) - I_2(s)] = \frac{100}{s}$$

$$I_1(s) [10 + 0.02s] - 0.02s I_2(s) = \frac{100}{s} \quad \text{--- (3)}$$

For eqn (2)

$$5I_2(s) + 0.02 [sI_2(s) - sI_1(s)] = 0$$

$$I_2(s) [5 + 0.02s] = 0.02s I_1(s)$$

$$I_2(s) = \frac{0.02s}{5 + 0.02s} I_1(s)$$

$$= \frac{s}{s + 250} I_1(s) \quad \text{--- (4)}$$

Substituting value of $I_2(s)$ in equation (3), we get

$$I_1(s) [10 + 0.02s] - 0.02s \cdot \frac{sI_1(s)}{s + 250} = \frac{100}{s}$$

$$I_1(s) \left[10 + 0.02s - \frac{0.02s^2}{s+250} \right] = \frac{100}{s}$$

$$I_1(s) \left[10(s+250) + 0.02s(s+250) - 0.02s^2 \right] = \frac{100}{s} (s+250)$$

$$I_1(s) \left[10s + 2500 + 0.02s^2 + 5s - 0.02s^2 \right] = \frac{100}{s} (s+250)$$

$$I_1(s) \left[15s + 2500 \right] = \frac{100}{s} (s+250)$$

$$I_1(s) = \frac{100(s+250)}{s \times 15(s+166.7)}$$

$$= 6.67 \left(\frac{s+250}{s(s+166.7)} \right)$$

$$\frac{s+250}{s(s+166.7)} = \frac{A}{s} + \frac{B}{(s+166.7)}$$

$$A = \frac{250}{166.7} = 1.5 \quad \& \quad B = -0.5$$

$$\therefore I_1(s) = \frac{6.67 \times 1.5}{s} - \frac{0.5 \times 6.67}{s+166.7}$$

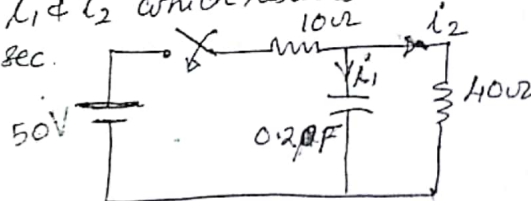
$$= \frac{10}{s} - \frac{3.33}{(s+166.7)}$$

$$i_1(t) = 10 - 3.33 e^{-166.7 t} \text{ Amp.}$$

$$I_2(s) = 6.67 \frac{(s+250)}{s(s+166.7)} \times \frac{s}{(s+250)} = \frac{6.67}{s+166.7}$$

$$i_2(t) = 6.67 e^{-166.7 t} \text{ Amp.}$$

⑤ In the two mesh network, there is no charge on the capacitor. Find the branch currents i_1 & i_2 which results when the switch is closed at $t=0$ sec.



The equations for the circuit are

$$10(i_1 + i_2) + \frac{1}{0.2} \int i_1 dt = 50 \quad \dots (1)$$

$$40i_2 - \frac{1}{0.2} \int i_1 dt = 0 \quad \dots (2)$$

Taking Laplace transforms

$$2I_1(s) + 2I_2(s) + \frac{I_1(s)}{s} = \frac{10}{s}$$

$$I_1(s) \left(2 + \frac{1}{s}\right) + 2I_2(s) = \frac{10}{s}$$

$$I_1(s) [1 + 2s] + 2sI_2(s) = 10 \quad \dots (3)$$

$$40I_2(s) - \frac{I_1(s)}{0.2s} = 0$$

$$40I_2(s) = \frac{I_1(s)}{0.2s}$$

$$I_2(s) = \frac{I_1(s)}{8s} \quad \dots (4)$$

Substituting $I_2(s)$ in (3), we get

$$(1 + 2s)I_1(s) + 2s \cdot \frac{I_1(s)}{8s} = 10$$

$$(1 + 8s)I_1(s) + I_1(s) = 40$$

$$I_1(s) [5 + 8s] = 40$$

$$I_1(s) = \frac{40}{(8s + 5)} = \frac{5}{s + 5/8}$$

$$i_1(t) = \underline{\underline{5e^{-5/8t} \text{ Amp}}}$$

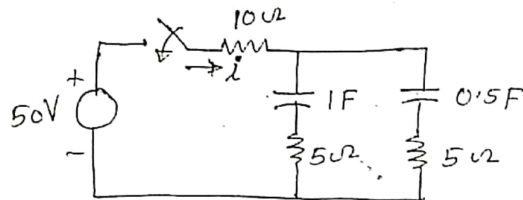
$$\begin{aligned} I_2(s) &= \frac{1}{8s} \cdot \frac{5}{(s + 5/8)} = \frac{5/8}{s(s + 5/8)} \\ &= \frac{A}{s} + \frac{B}{(s + 5/8)} \end{aligned}$$

on solving, $A = 1, B = -1$

$$I_2(s) = \frac{1}{s} - \frac{1}{s + 5/8}$$

$$\therefore i_2(t) = \underline{\underline{(1 - e^{-5/8t}) \text{ Amp}}}$$

- (5) In the network shown below, the switch is closed at $t=0$ sec and there is no initial ~~current~~ charge on either capacitors. Find the resultant current i . Draw the transformed circuit (s domain)

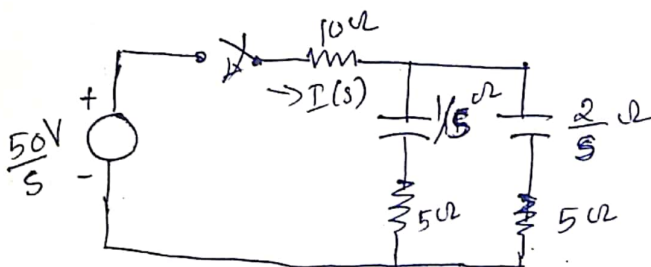


$$Z(s) = 10 + \frac{10 + (5 + 1/s)(5 + \frac{1}{0.5s})}{10 + \frac{1}{s} + \frac{1}{0.5s}}$$

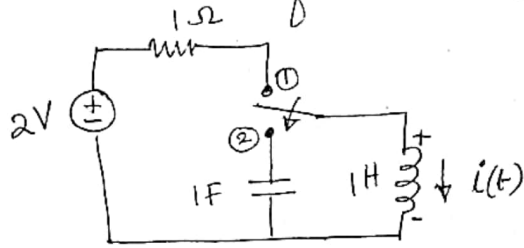
$$\text{Then } I(s) = \frac{V(s)}{Z(s)} = \frac{50/s}{10 + \frac{(5 + 1/s)(5 + \frac{1}{0.5s})}{(10 + 1/s + \frac{1}{0.5s})}}$$

$$i(t) = \mathcal{L}^{-1}[I(s)]$$

Transformed circuit



- ① In the circuit shown in figure, steady state exists when switch is in position 1. At time $t = 0$, it is moved to position 2. Determine the expression for current $i(t)$ through inductance for $t \geq 0$. (KTU, July 2017)



$$i(0^-) = \frac{2}{1} = 2A$$

At $t = 0^+$

$$\frac{1}{C} \int i dt + L \frac{di}{dt} = 0$$

$$\frac{1}{Cs} I(s) + L[sI(s) - i(0)] = 0$$

$$\frac{1}{s} I(s) + sI(s) - 2 = 0$$

$$I(s) \left[\frac{1}{s} + s \right] = 2$$

$$I(s) = \frac{2s}{s^2 + 1}$$

$$i(t) = 2 \cdot \mathcal{L}^{-1} \left[\frac{s}{s^2 + 1} \right] = \underline{\underline{2 \cdot \cos t}}$$

- ② The current through a 4F capacitance is given by the following s-domain equation $I(s) = \frac{24(s+2)}{(s+1)(s+3)}$. Find the voltage across the capacitance $V(t)$. (KTU-July 2017)

$$I(s) = \frac{24(s+2)}{(s+1)(s+3)} = \frac{A}{s+1} + \frac{B}{s+3}$$

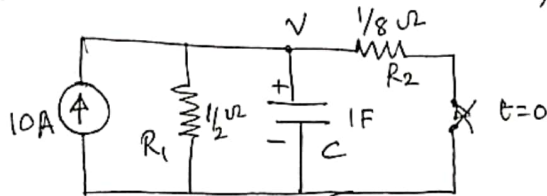
$$= \frac{12}{s+1} + \frac{12}{s+3}$$

$$\begin{cases} A = 12 \\ B = 12 \end{cases}$$

$$i(t) = \underline{\underline{(12e^{-t} + 12e^{-3t}) \text{ Amp}}}$$

$$\begin{aligned}
 V(t) &= \frac{1}{2} \int i(t) dt = \frac{1}{4} \int (12e^{-t} + 12e^{-3t}) dt = \frac{12}{4} \left(\int e^{-t} dt + \int e^{-3t} dt \right) \\
 &= 3 \left[\frac{e^{-t}}{-1} + \frac{e^{-3t}}{-3} \right]_0^t = \cancel{3e^{-t} - e^{-3t}} \\
 &= 3 \left[\frac{e^{-t}-1}{-1} + \frac{e^{-3t}-1}{-3} \right] = -3e^{-t} + 3 - e^{-3t} + 1 \\
 &= \underline{4 - 3e^{-t} - e^{-3t} \text{ Volts}}
 \end{aligned}$$

③ In the circuit shown below, the switch is



closed at $t=0$.

Find voltage across capacitor.

$$V(0) = 10 \times \frac{1}{2} = 5V$$

When the switch is closed

$$10 = \frac{V}{R_1} + C \frac{dV}{dt} + \frac{V}{R_2}$$

$$10 = \frac{V}{1/2} + 1 \frac{dV}{dt} + \frac{V}{1/8} = 2V + 8V + \frac{dV}{dt}$$

$$10V + \frac{dV}{dt} = 10 \quad \text{Taking Laplace transform}$$

$$10V(s) + 5V(s) - V(0) = \frac{10}{s}$$

$$10V(s) + 5V(s) - 5 = \frac{10}{s}$$

$$10V(s) + 5V(s) = \frac{10}{s} + 5$$

$$V(s) (10 + 5) = \frac{10}{s} + 5$$

$$V(s) = \frac{10 + 5s}{s(s+10)} = \frac{A}{s} + \frac{B}{s+10}$$

$$\begin{cases} A = 1 \\ B = 4 \end{cases}$$

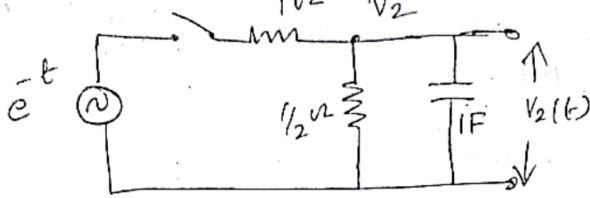
$$V(s) = \frac{1}{s} + \frac{4}{s+10}$$

$$V(t) = \underline{(1 + 4e^{-10t}) \text{ Volts}}$$

④ In the circuit shown, the switch is closed at $t=0$.

At $t=0$, the capacitor has an initial voltage of 0.5 Volts.

Find $V_2(t)$.



$$\frac{e^{-t} - V_2}{1} = \frac{V_2}{1/2} + 1 \frac{dV_2}{dt}$$

Taking LT

$$\frac{1}{s+1} - V_2(s) = 2V_2(s) + sV_2(s) - V_2(0)$$

$$\frac{1}{s+1} - V_2(s) = 2V_2(s) + sV_2(s) - 0.5$$

$$\frac{1}{s+1} + 0.5 = 3V_2(s) + sV_2(s)$$

$$= V_2(s) (3 + s)$$

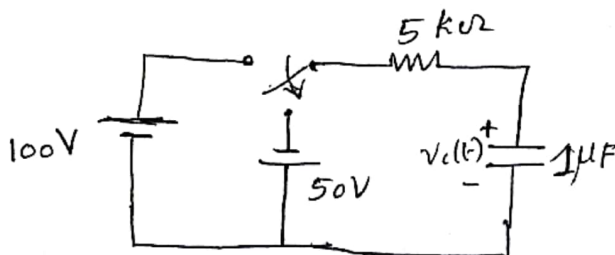
$$1 + 0.5(s+1) = V_2(s) (s+1) (s+3)$$

$$V_2(s) = \frac{0.5s + 1.5}{(s+1)(s+3)} = \frac{0.5(s+3)}{(s+1)(s+3)} = \frac{0.5}{s+1}$$

$$V_2(t) = \underline{\underline{0.5 e^{-t} \text{ Volts}}}$$

Example

The switch in the circuit is moved from position ① to ② at $t = 0$ sec. Determine $V_c(t)$



$$V_0 = -100V$$

Applying KVL

$$-50 - 5000 i' - 10^6 \int i' dt - 100 = 0$$

$$5000 i' + 10^6 \int i' dt = -150$$

$$5000 I(s) + 10^6 \frac{I(s)}{s} = -\frac{150}{s}$$

$$I(s) \left[5000 + \frac{10^6}{s} \right] = -\frac{150}{s}$$

$$I(s) = \frac{-150}{s(5000s + 10^6)} = \frac{-150}{5000s + 10^6}$$

taking ILT

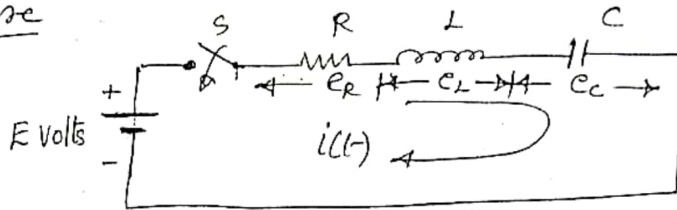
$$= \frac{-150}{5000(s + \frac{10^6}{5000})} = \frac{-150}{5000(s + 200)}$$

$$i(t) = \frac{-15}{500} e^{-200t} \text{ Amp}$$

$$\begin{aligned} v_c(t) &= \frac{1}{C} \int i(t) dt = 10^6 \times \frac{-15}{500} \frac{e^{-200t}}{-200} = \frac{15 \times 10^6}{10^5} e^{-200t} \\ &= \underline{\underline{150 e^{-200t} \text{ V}}} \end{aligned}$$

Transients in RLC series circuit (DC transients)

Step response



Let us assume that there is no initial current and no initial charge on the capacitor. Applying KVL, after closing switch at $t = 0$ sec

$$Ri + L \frac{di}{dt} + \frac{1}{C} \int i dt = E$$

Taking Laplace transforms on both sides

$$R I(s) + L [s I(s) - i(0^+)] + \frac{1}{Cs} I(s) = E/s$$

$$I(s) \left[R + Ls + \frac{1}{Cs} \right] = E/s$$

$$\begin{aligned} I(s) &= \frac{E/s}{R + Ls + 1/Cs} = \frac{E}{s(R + Ls + 1/Cs)} = \frac{E}{Rs + Ls^2 + 1/C} \\ &= \frac{E/L}{s^2 + R/Ls + 1/LC} \end{aligned}$$

Then $i(t)$ can be obtained by taking inverse transform.

For that find the factors of $s^2 + R/Ls + 1/LC$

$$\begin{aligned} s &= \frac{-R/L \pm \sqrt{(R/L)^2 - 4/LC}}{2} = \frac{-R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 - 1/LC} \\ &= \alpha \pm \beta \end{aligned}$$

$$\therefore I(s) = \frac{E/L}{[s - (\alpha + \beta)][s - (\alpha - \beta)]}, \text{ where}$$

$$\alpha = -\frac{R}{2L} \quad \text{and} \quad \beta = \sqrt{\left(\frac{R}{2L}\right)^2 - 1/LC}$$

Looking at the discriminant, it is clear that, there are three possibilities.

Case(a) discriminant is positive

$$\text{i.e. } \left(\frac{R}{2L}\right)^2 > \frac{1}{LC} \quad \text{or } R > 2\sqrt{L/C}$$

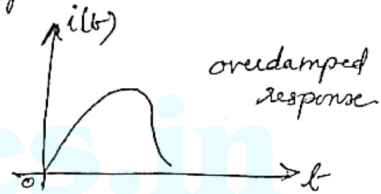
Then the two roots are real and distinct, and these roots are $(\alpha + \beta)$ and $(\alpha - \beta)$

$$\text{Then } I(s) = \frac{E/L}{(s - (\alpha + \beta))(s - (\alpha - \beta))} = \frac{A}{s - (\alpha + \beta)} + \frac{B}{s - (\alpha - \beta)}$$

A & B can be determined as before and we will get the solution for i as

$$\begin{aligned} i(t) &= A e^{(\alpha + \beta)t} + B e^{(\alpha - \beta)t} \\ &= e^{\alpha t} [A e^{\beta t} + B e^{-\beta t}] = e^{-R/2L t} [A e^{\beta t} + B e^{-\beta t}] \end{aligned}$$

In this case, the response is nonoscillatory and is said to be overdamped.



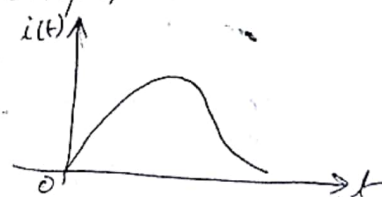
Case(b) Discriminant is zero

$$\text{i.e. } \left(\frac{R}{2L}\right)^2 = \frac{1}{LC} \quad \text{or } R = 2\sqrt{L/C}$$

In this case the two roots are equal and

$$\begin{aligned} I(s) &= \frac{E/L}{(s - \alpha)^2} = \frac{A}{(s - \alpha)^2} + \frac{B}{(s - \alpha)} \\ &= A t e^{\alpha t} + B e^{\alpha t} \\ &= e^{\alpha t} [A t + B] = e^{-R/2L t} [A t + B] \end{aligned}$$

The current is said to be critically damped



Case (c) Discriminant is negative

$$\text{ie when } \left(\frac{R}{2L}\right)^2 < \frac{1}{LC} \quad \text{or } R < 2\sqrt{L/C}$$

then the two roots become complex conjugates and these two roots are $(\alpha + j\beta)$ and $(\alpha - j\beta)$. Then

$$I(s) = \frac{A}{s - (\alpha + j\beta)} + \frac{B}{s - (\alpha - j\beta)} \quad \Phi$$

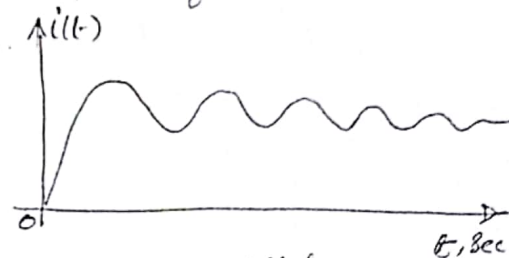
$$\begin{aligned} i(t) &= A e^{(\alpha + j\beta)t} + B e^{(\alpha - j\beta)t} \\ &= e^{\alpha t} [A e^{j\beta t} + B e^{-j\beta t}] \\ &= e^{\alpha t} [A (\cos \beta t + j \sin \beta t) + B (\cos \beta t - j \sin \beta t)] \end{aligned}$$

Here A & B are complex conjugates.

$$\begin{aligned} i(t) &= e^{\alpha t} [\cos \beta t (A+B) + j \sin \beta t (A-B)] \\ &= e^{\alpha t} [C \cos \beta t + D \sin \beta t] \\ &= e^{\alpha t} [K \sin(\beta t + \phi)] \end{aligned}$$

This solution shows that the current is oscillatory and at the same time decays in a short period of time as $\alpha = -R/2L$ is always negative.

This response is underdamped



The exponential term

$e^{\alpha t} = e^{-R/2L t}$ accounts for the decay of oscillations and is called decay factor or damping factor.

It makes each current peak a definite fraction less than that preceding it.

Now if $R=0$, $\beta = \frac{1}{\sqrt{LC}}$ and the response is

$$I(s) = \frac{E/L}{s^2 + 1/LC} = \frac{E/L}{(s + j/\sqrt{LC})(s - j/\sqrt{LC})} = \frac{A}{s + j/\sqrt{LC}} + \frac{B}{s - j/\sqrt{LC}}$$

$$\text{and } i(t) = A e^{-\frac{1}{2\sqrt{LC}}t} + B e^{\frac{1}{2\sqrt{LC}}t}$$

In this case $i(t)$ is undamped, i.e. oscillations will not die out.

Thus the value of resistance, determines the nature of transient current.

Among these three cases, the underdamped case is the most interesting as well as most frequently encountered case.

Damping is the process of reducing the amplitude of oscillations. Damping is the characteristics of the resistance parameter to dissipate energy. Consequently, its presence serves to prevent an uninterrupted interchange of energy between the two energy storing elements. Such an uninterrupted interchange of energy will constitute a sustained or undamped oscillations.

To describe the response of underdamped system, 2 figures of merit are used. They are damping ratio & natural frequency.

$$\text{Damping ratio, } \xi = \frac{\text{Actual value of } R}{\text{'R' for critical damping}} = \frac{R}{2\sqrt{L/C}}$$

$$\text{natural frequency, } \omega_n = \frac{1}{\sqrt{LC}}$$

$$\text{Then } \frac{R}{2L} = \frac{\xi \cdot 2\sqrt{L/C}}{2L} = \xi \sqrt{1/LC} = \xi \omega_n$$

$$\text{Also, for underdamped case, } \beta = \sqrt{\frac{1}{LC} - \left(\frac{R}{2L}\right)^2}$$

$$\text{Roots are complex conjugates and are } \alpha \pm j\beta = \frac{-R}{2L} \pm j\sqrt{\frac{1}{LC} - \left(\frac{R}{2L}\right)^2}$$

$$\therefore \beta = \sqrt{\frac{1}{LC} - \left(\frac{R}{2L}\right)^2} = \sqrt{\omega_n^2 - \xi^2 \omega_n^2}$$

$$\therefore \omega_d = \omega_n \sqrt{1 - \xi^2} \quad \text{is called}$$

Undamped frequency of oscillations, $\omega_d = \omega_n \sqrt{1 - \xi^2}$

Then, summarizing the nature of responses
when $R=0$, response is undamped i.e oscillatory.
when $R < 2\sqrt{L/C}$, response is underdamped
when $R = 2\sqrt{L/C}$, the response is critically damped.
when $R > 2\sqrt{L/C}$, the response is overdamped.

Problems :

- ① A series RLC circuit with $R=100\Omega$, $L=0.1H$ and $C=100\mu F$ has a DC voltage of 200V applied to it at $t=0$ through a switch. Find the expression for the transient current. Assume initially relaxed circuit conditions (i.e. $i_0=0$ & $V_C=0$)

$$Ri + L \frac{di}{dt} + \frac{1}{C} \int i dt = E$$

$$100i + 0.1 \frac{di}{dt} + \frac{1}{100 \times 10^{-6}} \int i dt = 200$$

$$100I(s) + 0.1 [sI(s) - i(0^+)] + \frac{10^6}{100s} I(s) = \frac{200}{s}$$

$$100I(s) + 0.1 sI(s) + \frac{10^4}{s} I(s) = \frac{200}{s}$$

$$I(s) \left[100 + 0.1s + \frac{10^4}{s} \right] = \frac{200}{s}$$

$$I(s) = \frac{200}{s(100 + 0.1s + 10^4/s)} = \frac{200}{100s + 0.1s^2 + 10^4} = \frac{200}{0.1(s^2 + 1000s + 10^5)}$$

$$= \frac{2000}{s^2 + 1000s + 10^5}$$

Roots of the denominator are $s_{1,2} = \frac{-1000 \pm \sqrt{1000^2 - 4 \times 10^5}}{2}$

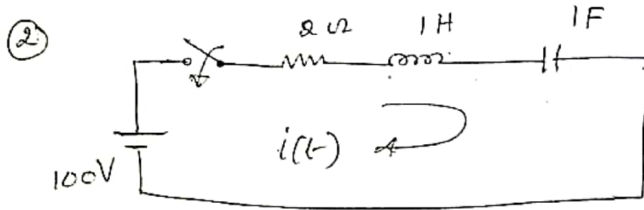
$$s_1, s_2 = -112.7 \text{ \& } -887.3$$

$$\therefore I(s) = \frac{2000}{(s+112.7)(s+887.3)} = \frac{A}{(s+112.7)} + \frac{B}{(s+887.3)}$$

On solving, $A = -B = 2.582$

$$\therefore I(s) = \frac{2.582}{s+112.7} - \frac{2.582}{s+887.3}$$

$$i(t) = (2.582 e^{-112.7t} - 2.582 e^{-887.3t}) \text{ Amp}$$



Determine the current in the circuit when the switch is closed at $t=0$ sec. Assume $i(0)=0$ & $q(0)=0$

Soln:

$$2i + \frac{di}{dt} + \int i dt = 100$$

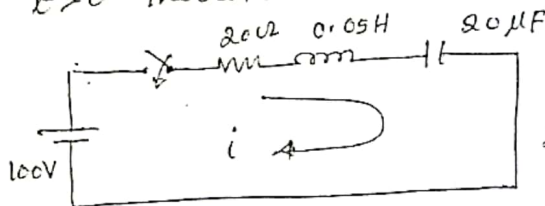
$$2I(s) + sI(s) + \frac{I(s)}{s} = \frac{100}{s}$$

$$I(s) \left[2 + s + \frac{1}{s} \right] = \frac{100}{s}$$

$$I(s) = \frac{100}{(s^2 + 2s + 1)} = \frac{100}{(s+1)^2}$$

$$i(t) = 100te^{-t} \text{ Amp}$$

③ For the circuit shown, determine the current at any time $t > 0$. The switch is closed at $t=0$. Assume zero initial conditions.



$$20i + 0.05 \frac{di}{dt} + \frac{10^6}{20} \int i dt = 100$$

$$20I(s) + 0.05 sI(s) + \frac{10^6}{20s} I(s) = \frac{100}{s}$$

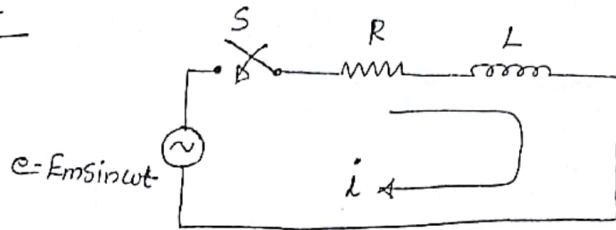
$$I(s) \left[20 + 0.05s + \frac{10^6}{20s} \right] = \frac{100}{s}$$

$$I(s) = \frac{100}{s \left(20 + 0.05s + \frac{10^6}{20s} \right)} \quad \text{On solving}$$

$$i(t) = e^{-200t} (2.04 \sin 979.8t) \text{ Amp}$$

AC Transients (Transient response to sinusoidal inputs)

RL circuit



When a sinusoidal voltage $E_m \sin \omega t$ is applied to the series R-L circuit at $t=0$ through a switch. (Assume zero initial conditions)

$$Ri + L \frac{di}{dt} = E_m \sin \omega t$$

Taking Laplace transform on both sides of the equation,

$$RI(s) + L[SI(s) - i(0^+)] = E_m \cdot \frac{\omega}{s^2 + \omega^2}$$

$$I(s) [R + Ls] = \frac{E_m \cdot \omega}{s^2 + \omega^2}$$

$$I(s) = \frac{E_m \cdot \omega}{(s^2 + \omega^2)(R + Ls)} = \frac{E_m \cdot \omega}{L(s^2 + \omega^2)(s + R/L)}$$

$$= \frac{E_m \cdot \omega / L}{(s^2 + \omega^2)(s + R/L)} = \frac{E_m \cdot \omega / L}{(s + R/L)(s + j\omega)(s - j\omega)}$$

$$= \frac{A}{(s + R/L)} + \frac{B}{(s + j\omega)} + \frac{C}{(s - j\omega)}$$

Solving for A, B & C , we get-

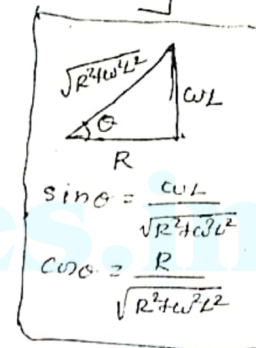
$$A = \frac{E_m \cdot \omega}{L(R^2 + \omega^2 L^2)} = \frac{E_m \cdot \omega L}{R^2 + \omega^2 L^2}$$

$$B = \frac{-E_m(\omega L - jR)}{2(R^2 + \omega^2 L^2)}$$

$$C = B^* = \frac{-E_m(\omega L + jR)}{2(R^2 + \omega^2 L^2)}$$

$$\therefore i(t) = \frac{E_m \omega L}{R^2 + \omega^2 L^2} e^{-R/L t} - \frac{E_m}{2(R^2 + \omega^2 L^2)} \left[(\omega L - jR) e^{-j\omega t} + (\omega L + jR) e^{j\omega t} \right]$$

$$\begin{aligned}
 i(t) &= \frac{\omega L \cdot E_m}{R^2 + \omega^2 L^2} \left[e^{-R/L t} \frac{(e^{-j\omega t} + e^{j\omega t})}{2} - \frac{jR}{\omega L} \frac{(e^{j\omega t} - e^{-j\omega t})}{2} \right] \\
 &= \frac{\omega L \cdot E_m}{R^2 + \omega^2 L^2} \left[e^{-R/L t} \cos \omega t - \frac{j^2 R}{\omega L} \frac{(e^{j\omega t} - e^{-j\omega t})}{2j} \right] \\
 &= \frac{\omega L \cdot E_m}{R^2 + \omega^2 L^2} \left[e^{-R/L t} \cos \omega t + \frac{R}{\omega L} \sin \omega t \right] \\
 &= \frac{E_m}{\sqrt{R^2 + \omega^2 L^2}} \left[\frac{\omega L}{\sqrt{R^2 + \omega^2 L^2}} e^{-R/L t} \cos \omega t + \frac{R}{\sqrt{R^2 + \omega^2 L^2}} \sin \omega t \right] \\
 &= \frac{E_m}{\sqrt{R^2 + \omega^2 L^2}} \left[e^{-R/L t} \sin \theta - \sin \theta \cos \omega t + \cos \theta \sin \omega t \right] \\
 &= \frac{E_m}{\sqrt{R^2 + \omega^2 L^2}} \left[e^{-R/L t} \sin \theta + \sin(\omega t - \theta) \right]
 \end{aligned}$$



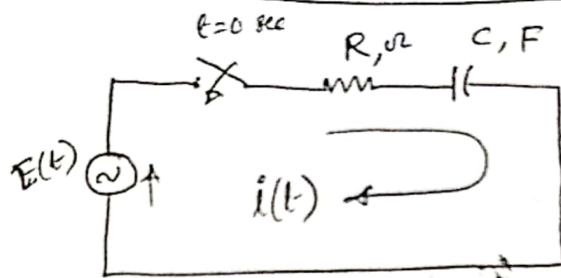
In a similar way RC & RLC circuit response can be found out.

Note: In deriving the above equation, the sinusoidal function has been taken as $E_m \sin \omega t$ which is zero at $t=0$. Actually the voltage function can be at any point of the wave at $t=0$ i.e. at the instant of closing the switch. To include this effect, the voltage function can be written as $E_m \sin(\omega t + \phi)$, where ϕ can be any value from 0 to 2π .

In that case, the above equation can be written as

$$i(t) = \frac{E_m}{\sqrt{R^2 + \omega^2 L^2}} \left[\sin(\omega t + \phi - \theta) + e^{-R/L t} \cdot \sin(\phi - \theta) \right]$$

R-C circuit response to sinusoidal excitation



Assume that initial charge on the capacitor is zero.
The KVL equation is

$$Ri + \frac{1}{C} \int i dt = E_m \sin \omega t$$

Taking Laplace transform, we get

$$I(s) \left[R + \frac{1}{Cs} \right] = E_m \cdot \left(\frac{\omega}{s^2 + \omega^2} \right)$$

$$I(s) = \frac{\omega \cdot E_m}{(s^2 + \omega^2) \left(R + \frac{1}{Cs} \right)} = \frac{\omega E_m Cs}{(s^2 + \omega^2) \left(s + \frac{1}{RC} \right)}$$

$$I(s) = \frac{\omega E_m}{R} \left[\frac{s}{(s^2 + \omega^2) \left(s + \frac{1}{RC} \right)} \right]$$

$$I(s) = \frac{A}{s + \frac{1}{RC}} + \frac{B}{s + j\omega} + \frac{C}{s - j\omega}$$

$$i(t) = A e^{-\frac{t}{RC}} + B e^{-j\omega t} + C e^{j\omega t} \quad \text{where}$$

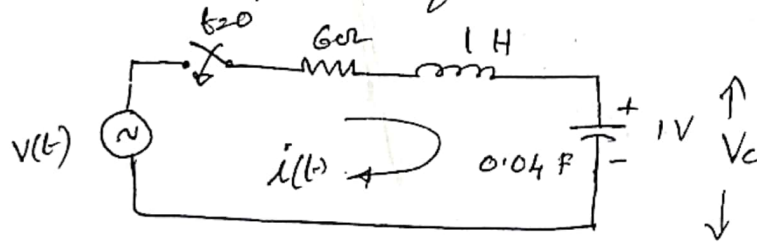
$$A = \frac{-\omega C V_m}{1 + \omega^2 R^2 C^2}, \quad B = \frac{\omega C V_m (1 - j\omega RC)}{2(1 + \omega^2 R^2 C^2)}$$

$$C = \frac{\omega C E_m (1 + j\omega RC)}{2(1 + \omega^2 R^2 C^2)}$$

On simplifying, we get

$$i(t) = \frac{E_m}{\sqrt{R^2 + \frac{1}{\omega^2 C^2}}} \sin(\omega t + \theta) - \frac{E_m}{\sqrt{R^2 + \frac{1}{\omega^2 C^2}}} e^{-t/RC}$$

A series R-L-C circuit excited by a voltage $V(t) = 12 \sin 5t$.
 The initial current in the circuit is 5 A and the initial voltage across the capacitor is 1 volt with polarity shown in fig.
 Find the expression for the current.



KVL equation in time domain is

$$60i + \frac{di}{dt} + \frac{1}{0.04} \int i dt = 12 \sin 5t$$

Taking Laplace transforms

$$60I(s) + sI(s) - 5 + 25 \frac{I(s)}{s} + \frac{1}{s} = \frac{12 \times 5}{s^2 + 5^2}$$

$$I(s) \left[60 + s + \frac{25}{s} \right] = \frac{60}{s^2 + 25} + 5 - \frac{1}{s}$$

$$I(s) = \frac{60s}{(s^2 + 25)(s^2 + 6s + 25)} + \frac{5s - 1}{s^2 + 6s + 25}$$

$$= \frac{j1.25}{(s + 3 - j4)} - \frac{j1.25}{s + 3 + j4} + \frac{j}{s + j5} - \frac{j}{s - j5} \\ + \frac{2.5 + j2}{s + 3 - j4} + \frac{2.5 - j2}{s + 3 + j4}$$

Taking inverse transform, we get after simplification

$$i(t) = (5e^{-3t} \cos 4t - 6.5e^{-3t} \sin 4t + 2 \sin 5t) \text{ amp}$$